

PRACTICAL 2: Professor Evaluation and Beauty Dataset

Submitted By: Madhvi Sharma

1. Dataset Introduction

The **Professor Evaluation and Beauty Dataset** contains information related to professor/course evaluations. The dataset consists of **463 observations from 94 professors** at the University of Texas at Austin.

The dataset contains information such as teaching evaluation score, professor age, gender, academic rank, number of students in the class, and average beauty score.

Dataset Type

The dataset is **cross-sectional data** because the observations represent different professor/course evaluations rather than observations collected over a period of time.

2. Python Implementation

```
[1]: # PROFESSOR EVALUATION AND BEAUTY DATASET

# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# LOAD DATASET
df = pd.read_csv("C:/Users/Ajsha/Downloads/evals.csv")

print("Dataset loaded successfully!")
print("Number of observations:", len(df))

print("\nFirst 5 rows:")
print(df.head())

# Q1. TIME SERIES OR CROSS-SECTIONAL?

print("\nQ1. TYPE OF DATA")

print("The teacher rating data is CROSS-SECTIONAL data.")
print("Reason: The dataset contains observations of different")
print("professors/courses and is not collected over time.")

# Q2. MEAN, MEDIAN, MINIMUM AND MAXIMUM OF STUDENTS

print("\nQ2. NUMBER OF STUDENTS")

print("Mean      :", round(df["cls_students"].mean(), 2))
print("Median    :", df["cls_students"].median())
print("Minimum    :", df["cls_students"].min())
print("Maximum    :", df["cls_students"].max())
```

```
# Q3. DESCRIPTIVE STATISTICS
```

```
print("\nQ3. DESCRIPTIVE STATISTICS")  
print(df.describe())
```

```
# Q4. HISTOGRAM OF BEAUTY VARIABLE
```

```
print("\nQ4. HISTOGRAM OF BEAUTY SCORE")  
plt.hist(  
    df["bty_avg"],  
    bins=10,  
    color="purple",  
    edgecolor="black"  
)  
  
plt.xlabel("Beauty Score")  
plt.ylabel("Frequency")  
plt.title("Histogram of Professor Beauty Scores")  
plt.show()
```

```
print("Comment:")  
print("The beauty score distribution is approximately unimodal.")  
print("Most observations are concentrated around the middle range.")
```

```
# Q5. BEAUTY SCORE BY GENDER
```

```
print("\nQ5. BEAUTY SCORE BY GENDER")  
result = df.groupby("gender")["bty_avg"].agg(["mean", "std"])  
print(result.round(3))
```

```
# Q6. TENURE STATUS BY GENDER
```

```
print("\nQ6. TENURE STATUS BY GENDER")  
  
df["Tenured"] = df["rank"] == "tenured"  
  
result = pd.crosstab(  
    df["gender"],  
    df["Tenured"],  
    normalize="index"  
) * 100  
  
print(result.round(2))
```

3. Results and Analysis :

Initial Dataset Output

```
Dataset loaded successfully!  
Number of observations: 463
```

```
First 5 rows:
```

```
   course_id  prof_id  score      rank  ethnicity  gender  language \  
0          1        1    4.7  tenure track   minority  female  english  
1          2        1    4.1  tenure track   minority  female  english  
2          3        1    3.9  tenure track   minority  female  english  
3          4        1    4.8  tenure track   minority  female  english  
4          5        2    4.6      tenured  not minority   male  english
```

```
   age  cls_perc_eval  cls_did_eval  ...  cls_credits  bty_f1lower  \  
0   36      55.81395          24  ...  multi credit          5  
1   36      68.80000          86  ...  multi credit          5  
2   36      60.80000          76  ...  multi credit          5  
3   36      62.60163          77  ...  multi credit          5  
4   59      85.00000          17  ...  multi credit          4
```

```
   bty_f1upper  bty_f2upper  bty_m1lower  bty_m1upper  bty_m2upper  bty_avg  \  
0             7             6             2             4             6      5.0  
1             7             6             2             4             6      5.0  
2             7             6             2             4             6      5.0  
3             7             6             2             4             6      5.0  
4             4             2             2             3             3      3.0
```

```
   pic_outfit  pic_color  
0  not formal    color  
1  not formal    color  
2  not formal    color  
3  not formal    color  
4  not formal    color
```

```
[5 rows x 23 columns]
```

Q1. Type of Data

The teacher rating data is cross-sectional data because the observations represent different professors/courses and are not collected over time.

Result

Q1. TYPE OF DATA

The teacher rating data is CROSS-SECTIONAL data.

Reason: The dataset contains observations of different professors/courses and is not collected over time.

Q2. Number of Students

Interpretation

The average number of students per class is approximately 55, while the median is 29. The smallest class has 8 students, whereas the largest class has 581 students.

Q2. NUMBER OF STUDENTS

```
Mean      : 55.18
Median    : 29.0
Minimum   : 8
Maximum   : 581
```

Q3. Descriptive Statistics

The describe() function was used to obtain the descriptive statistics of the numerical variables.

Q3. DESCRIPTIVE STATISTICS

```
count      course_id      prof_id      score      age      cls_perc_eval  \
mean      232.000000      45.153348      4.174730      48.365011      74.427788
std        133.800847      27.547052      0.543865      9.802742      16.756311
min         1.000000       1.000000      2.300000      29.000000      10.416670
25%        116.500000      20.000000      3.800000      42.000000      62.696165
50%        232.000000      43.000000      4.300000      48.000000      76.923080
75%        347.500000      70.500000      4.600000      57.000000      87.249170
max         463.000000      94.000000      5.000000      73.000000     100.000000

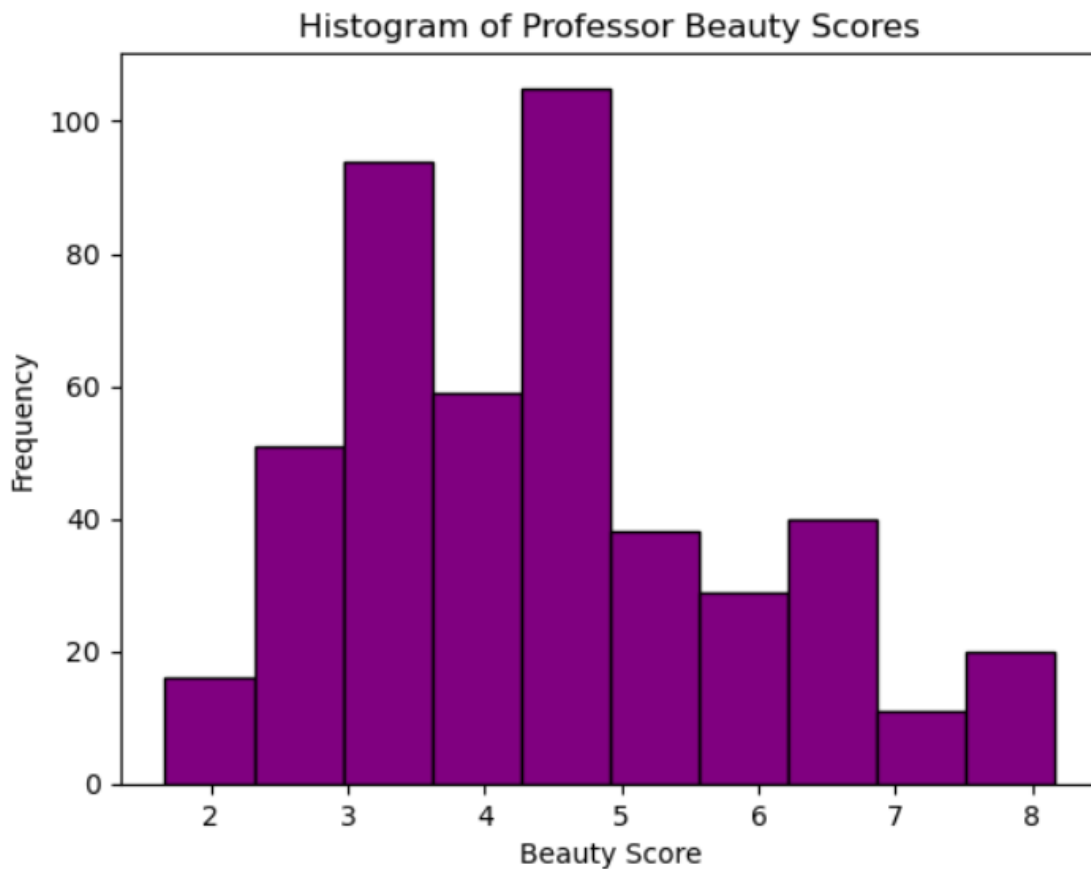
count      cls_did_eval      cls_students      bty_f1lower      bty_f1upper      bty_f2upper  \
mean        36.624190       55.177106       3.963283       5.019438       5.213823
std         45.018481       75.072800       1.873936       1.934437       2.018224
min          5.000000        8.000000       1.000000       1.000000       1.000000
25%         15.000000       19.000000       2.000000       4.000000       4.000000
50%         23.000000       29.000000       4.000000       5.000000       5.000000
75%         40.000000       60.000000       5.000000       7.000000       6.000000
max         380.000000      581.000000       8.000000       9.000000      10.000000

count      bty_m1lower      bty_m1upper      bty_m2upper      bty_avg
mean        3.412527       4.146868       4.751620       4.417844
std         1.637102       2.110586       1.575266       1.527380
min          1.000000       1.000000       1.000000       1.667000
25%          2.000000       3.000000       4.000000       3.167000
50%          3.000000       4.000000       5.000000       4.333000
75%          5.000000       5.000000       6.000000       5.500000
max          7.000000       9.000000       9.000000       8.167000
```

Q4. Histogram of Beauty Variable

The beauty score distribution is approximately unimodal. Most observations are concentrated around the middle range.

Q4. HISTOGRAM OF BEAUTY SCORE



Comment:

The beauty score distribution is approximately unimodal.
Most observations are concentrated around the middle range.

Q5. Average Beauty Score by Gender

Q5. BEAUTY SCORE BY GENDER

	mean	std
gender		
female	4.659	1.573
male	4.242	1.472

Conclusion

The average beauty score differs by gender descriptively. Female instructors have a higher average beauty score than male instructors.

Q6. Tenure Status by Gender

Aspect	Original	Corrected
Female tenured (%)	38.97 (evaluations)	38.97 (evaluations); 45.00 (professors)
Male tenured (%)	66.04 (evaluations)	66.04 (evaluations); 66.67 (professors)
"False" column	Unlabelled mix of tenure track + teaching	Explicitly shown as separate rank categories



Level of analysis	Course-level only	Both course-level and professor-level
Statistical claim	None (correctly descriptive)	Explicitly flagged as descriptive, not inferential

Conclusion

Approximately 38.97% of course evaluations for female instructors and 66.04% of course evaluations for male instructors involve tenured faculty. When analyzed at the professor level, 45.00% of female professors and 66.67% of male professors hold tenured rank. These are descriptive comparisons only; no statistical test was performed, so we cannot conclude that tenure status significantly differs by gender. However, the data are consistent with a gender gap in tenure, with a notably larger share of women still on the tenure track (35.38% vs. 14.55% of evaluations).

4. Overall Conclusion

The analysis of the Professor Evaluation and Beauty Dataset shows that the data is cross-sectional. The average class size is approximately 55 students, while the median class size is 29 students. The beauty score distribution is approximately unimodal. Female instructors have a higher average beauty score than male instructors, while the percentage of tenured male instructors is higher than that of female instructors.

